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Corolario del principio del módulo máximo

Corolario 1. *Sea* $\Omega \subset \mathbb{C}$ *un dominio acotado y* $f \in \mathcal{H}(\Omega) \cap \mathcal{C}\left(\overline{\Omega}\right)$

$$\implies \max_{z\in\overline{\Omega}} |f(z)| = \max_{z\in\partial\Omega} |f(z)| \, .$$

***Demostración*:** Como Ω es acotado, $\overline{\Omega}$ es compacto y, por tanto, como $f \in \mathcal{C}\left(\overline{\Omega}\right)$, $|f| \in \mathcal{C}\left(\overline{\Omega}\right)$ alcanza su máximo en $\overline{\Omega}$ (Teorema de Weierstrass). Veamos dos casos:

I. Si $\exists z_0 \in \Omega$ tal que $|f(z_0)| = \max_{z\in\overline{\Omega}} |f(z)|$, por el principio del módulo máximo, f es constante en Ω y, por tanto, $\max_{z\in\partial\Omega} |f(z)| = |f(z_0)| = \max_{z\in\overline{\Omega}} |f(z)|$.

II. Si $\forall z \in \Omega : |f(z)| < \max_{z\in\overline{\Omega}} |f(z)|$, entonces, necesariamente, $\exists z_1 \in \partial\Omega$ tal que $|f(z_1)| = \max_{z\in\overline{\Omega}} |f(z)|$. Por tanto, $\max_{z\in\partial\Omega} |f(z)| = |f(z_1)| = \max_{z\in\overline{\Omega}} |f(z)|$. ∎

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